Trend research — 2026-04-19 · incr 23

Flip timing is bimodal; |strength| >= 0.15 at bar 20 is a clean real-time gate; structure is 16x noisier than direction

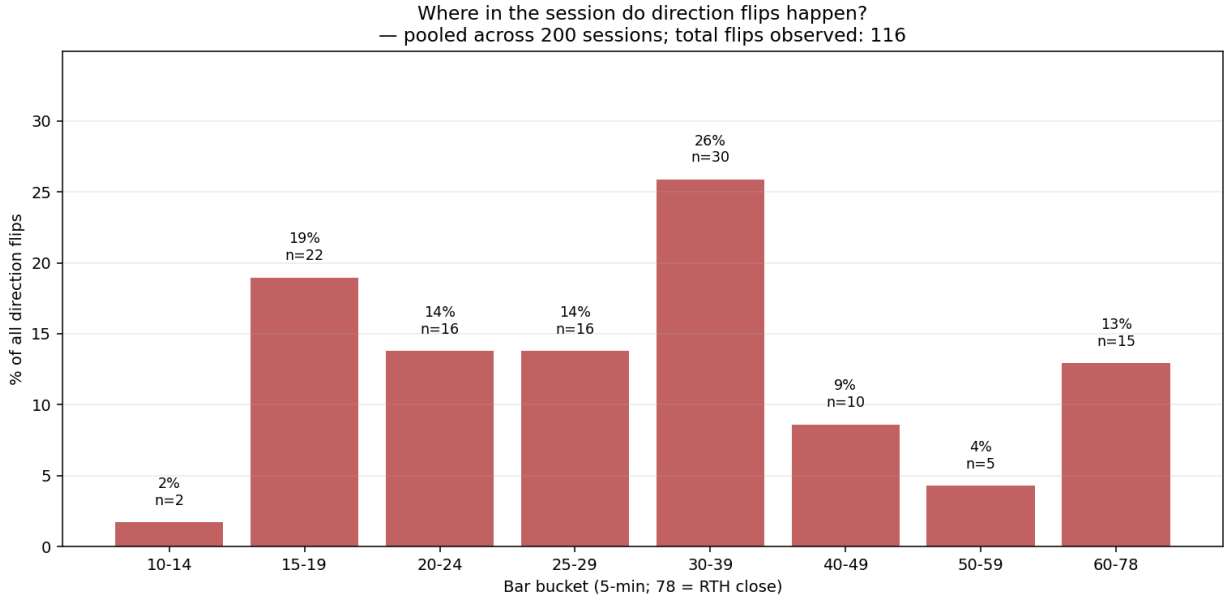
TL;DR

Incr 22 showed the live **TrendState.direction** rarely flips (63 % zero-flip rate, median lock-in bar 11). This increment answers three follow-up questions on **200 sessions**: (1) WHEN do flips happen? **64 %** fall in bars 15-39 — bimodal peaks at bars 15-19 and 30-39. (2) Does early strength predict stability? Yes — **|strength| ≥ 0.15 at bar 20** is a clean real-time gate: **93 %** direction-survival vs 47 % baseline, **85 %** zero-flip rate vs 46 % baseline. (3) Does **structure** stabilise with direction? No — **16× noisier** (1 905 structure flips vs 116 direction flips). But most structure movement is the intended spike → channel evolution. **No production change.**

Headline numbers

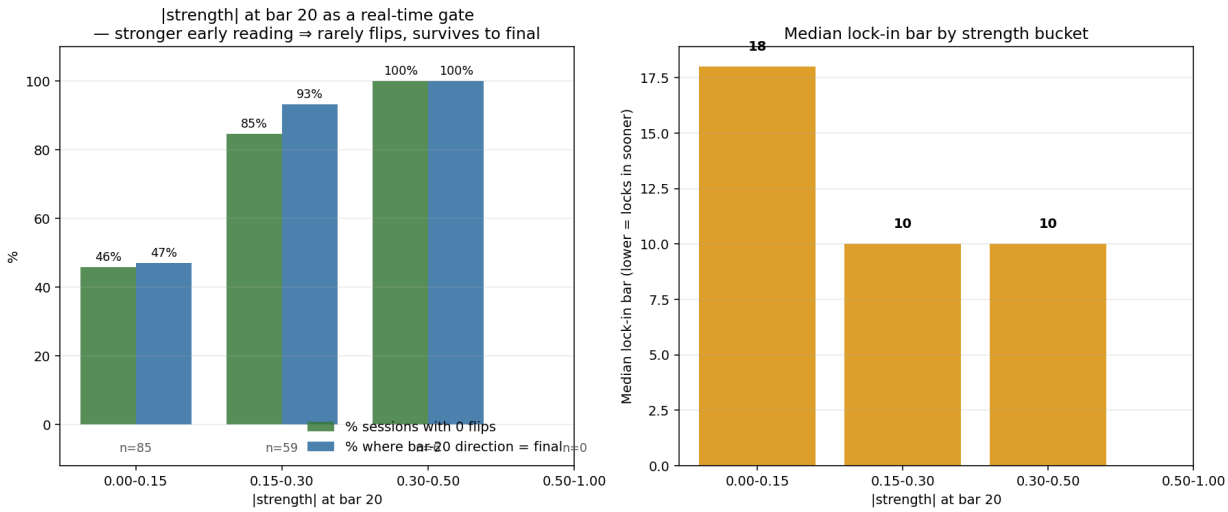
- Sample: **200 (symbol, session) pairs** from cache/databento/*.parquet, 5-min RTH. **12 129** per-bar trajectory rows persisted for reuse. Runtime: **355 s**.
- **Direction flips cluster in bars 15-39** — 64 % of all flips happen there. Bimodal peaks at bar 15-19 (**19 %**) and bar 30-39 (**26 %**). After bar 40, only **26 %** of flips occur. Waiting until bar 30 dodges most of the reversibility.
- **|strength| ≥ 0.15 at bar 20 is a bolt-on real-time gate**: **93 %** of these sessions see their bar-20 direction survive to the final bar (vs 47 % for |strength| < 0.15). **85 %** zero-flip rate (vs 46 %). Median lock-in drops from bar 18 → bar 10.
- **Structure is 16× noisier than direction**: 9.53 structure flips / session vs 0.58 direction flips. But 100 % → 0 % spike-share evolution between bar 10 and bar 30 is intended Brooks cycle-phase progression, not random flicker.
- **Reproduced incr 22's stability claim**: direction mean flips / session = 0.58 on 200 sessions vs 0.56 on incr 22's 300-session sample. Stability is real and reproducible.

Headline figure — WHERE flips happen (bimodal distribution)



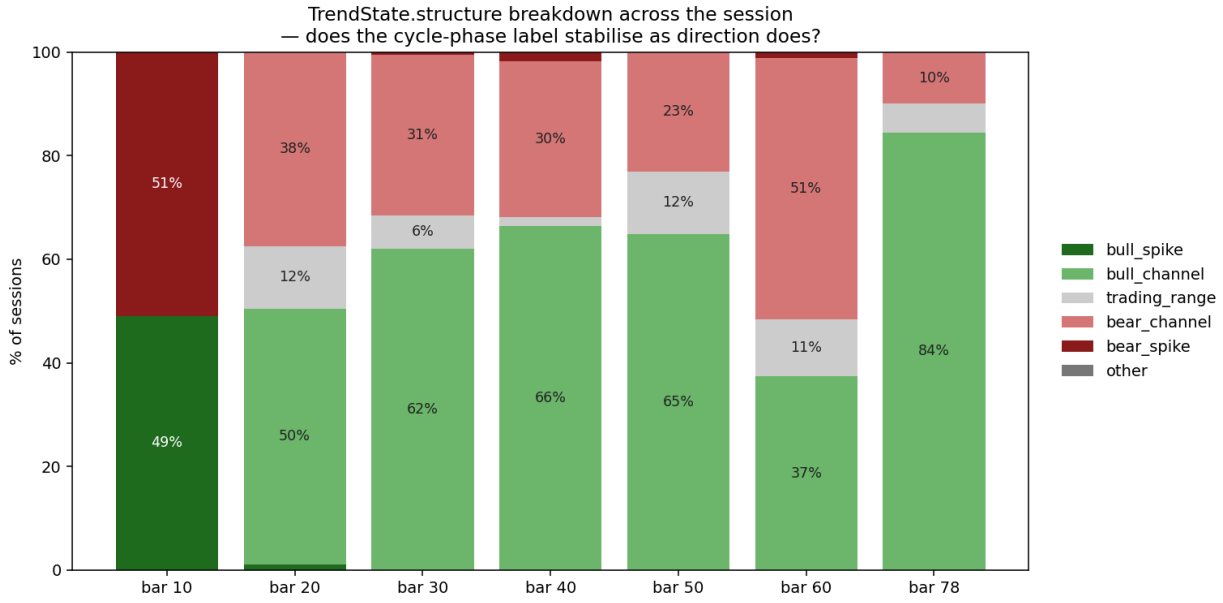
Percentage of all 116 observed direction flips falling in each bar bucket. The bar-15-19 bucket (19 %) and the bar-30-39 bucket (26 %) together hold 45 % of all flips. The 60-78 end-of-day bucket holds 13 % (per-bar rate is lower than bars 15-39). Implication: waiting for bar 30 reduces real-time reversal risk substantially, without giving up much live-reading usefulness.

The |strength| ≥ 0.15 gate — a clean real-time filter



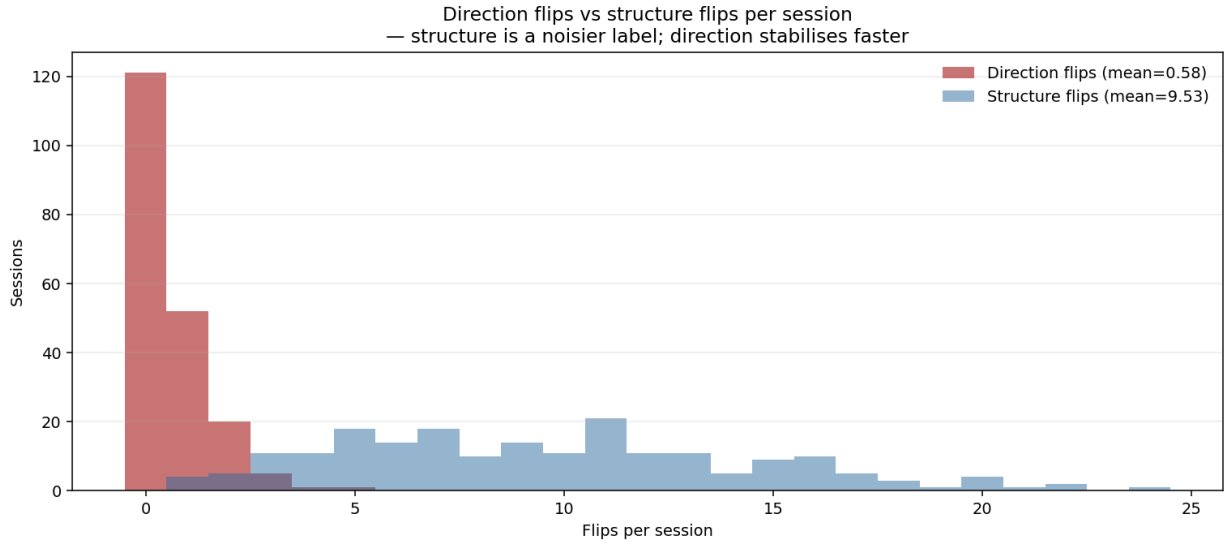
Left: sessions binned by |strength| at bar 20. Zero-flip share jumps from 46 % (weak readings) → 85 % (0.15-0.30) → 100 % (0.30-0.50). Direction-survival to final follows the same jump: 47 % → 93 % → 100 %. Right: median lock-in bar drops from 18 → 10 at the 0.15 threshold. The 0.50+ bucket is empty (n=0) because the equal-weighted 12-contributor aggregate damps early — not a ceiling, just a distribution note. Use this as a dashboard consumer rule, not a production threshold.

Structure trajectory — spike → channel evolution



At bar 10, 100 % of sessions are in opening-spike labels (49 % bull_spike, 51 % bear_spike). By bar 20 the spike share has collapsed to < 1 %; bull/bear channel dominates with trading_range at 12 %. By the end of the RTH subset (bar 78, n=90 full-day sessions), 94 % are in a channel label. This is intended Brooks cycle-phase evolution — NOT instability. Note: late-bar bars use the subset of sessions that reach that point (today's partial session truncates many).

Direction vs structure flip counts — 16× noise ratio



Histogram of per-session flip counts. Direction flips (red) are concentrated at 0; structure flips (blue) span 5-15. Headline: 1 905 structure flips vs 116 direction flips across the 200-session sample. Structure is intentionally a richer label space (6+ values vs 3) and captures ongoing cycle-phase evolution, so the higher flip count isn't a bug — but it means the front-end shouldn't treat structure changes as reversal signals.

What this means for live consumption

- **Real-time cadence is now explicit.** Bar 10: reading available but near-random. Bar 20 + |strength| ≥ 0.15: **high-confidence** (93 % dir-survival). Bar 30: 64 % of flipping sessions already had their reversal — reading is in the stable majority.
- **|strength| ≥ 0.15 at bar 20 is the gate the data is asking for.** It's a simple live rule: direction + strength together, not direction alone. Optional dashboard rule, not a scoring change.
- **Front-end shouldn't display structure with 'confidence'.** It's a discrete label telling a cycle-phase story; a flicker from bull_spike → bull_channel at bar 12 is normal, not noise.
- **Directional accuracy is still incr 19's problem.** This run measures self-consistency, not market-move prediction. The 93 % gate tells you which readings will survive — not whether they'll be right about the move.

Mistakes avoided

- **Didn't over-claim the low-strength bucket.** 47 % dir-survival for |strength| < 0.15 includes cases where bar-20 direction is 'none' and the final is 'up' — that's a late commit, not a flip. The zero-flip share (46 % vs 85 %) is the cleaner contrast.
- **Didn't claim a 0.50+ ceiling.** n=0 at |strength| ≥ 0.50 means the equal-weighted aggregate damps early, not that high-strength readings don't exist. The gate is at 0.15.
- **Didn't conflate structure flips with real instability.** 9.5 flips / session, but bar-10-to-bar-30 is 100 % → 0 % spike share — that's planned cycle-phase progression. Noted as flagged follow-up: separate 'intended evolution' flips from 'reversal' flips.
- **Didn't promote the 0.15 gate as a production change.** It's a dashboard consumer rule. Scoring, ranking, and the aggregate remain byte-stable.

What's next — requires Will's nod

- **Optional real-time consumer rule** — display 'high-confidence live direction' when |strength| ≥ 0.15 AND bar count ≥ 20. Front-end panel only; no change to compute_trend_state.
- **Still pending from earlier runs:** incr 21 (OPEN_EXTREME_THRESHOLD 0.15 → 0.25), incr 20 (ALWAYS_IN_WINDOW 5 → 10), incr 18 (MAJORITY_TREND_BAR_FLOOR 0.40 → 0.25), incr 19 (htf_alignment silent-zero docstring).
- **Future study** — structure-flip type taxonomy: split intended-evolution flips (bull_spike → bull_channel) from reversal flips (bull_channel → bear_channel). Likely 9 / session intended + 0.5 / session genuine-reversal split.
- **Future study** — day-type stratification: re-run with TFO / chop labels to check the aggregate isn't hiding bimodality between day types.

Findings site: https://github.com/zerosumsystems-ui/iPhone-/tree/main/trend-classification